

Review Questions Chapter #2

Q. Derive the formula for min-max normalization of data on $[1, -1]$ interval?

⇒ This technique scales the values of a feature to a range between 0 and 1.

⇒ This is done by subtracting the minimum value of the feature from each value, and then dividing by the range of the feature.

$$V' = \frac{V - \min(A)}{\max(A) - \min(A)} \cdot (\text{new_max}(A) - \text{new_min}(A)) + \text{new_min}(A)$$

Q. Given 1-D data set $X = \{-5.0, 23.0, 17.6, 7.23, 1.11\}$, normalize the dataset using:

- decimal scaling on interval
- min-max normalization on interval $[0, 1]$
- SD normalization.

Q. Characteristics of raw data?

Characteristics of raw data are as follows:

- 1) All raw data initially prepared for data mining are often large.
- 2) Data may be missing.
- 3) Data may be misrecording.
- 4) Data may not be from the population they are supposed to be from.

Q. Define time-dependent data?

- Time-dependent data also known as temporal data.
- Data that has specific time content and may vary over time.
- ex. customers address, product price.

Q. Define large dataset?

- Large dataset also known as big data.
- Consist of volumes of data so vast that traditional data processing software is inadequate to handle.

Q. What is missing data?

- most common and non-negotiable problem in statistical analysis & ML.
- They affect data quality.

◦ missing values are data points that are absent for a specific variable in a dataset

Types of MD

↓
Non-ignorable

↓
ignorable.